SIMATS ENGINEERING

CONTINUOUS INTERNAL ASSESSMENT

DATABASE MANAGEMENT SYSTEMS

Course Code: CSA0504

Student Name	Pugazharasan K
Register Number	192511341
Course Outcome	CO1 – DBMS architecture, data models, schema types, ER/EER design and relational schemas
Questions	05
Total Marks	100
Submission Date	05-10-26

Prepared as a complete, rubric-aligned submission draft

Declaration / Code of Conduct

I Pugazharasan K(192511341) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Student Signature: Pugazharasan K

Date: 05-10-2026

Assessment Scope and Outcome

This submission addresses the five questions in the supplied assessment brief for Database Management Systems (CSA05), mapped to CO1. The responses emphasize concept identification, correct linkages and cardinalities, organized application of EER/ER concepts, complete coverage of task requirements, and clear presentation. The assessment brief states that CO1 covers DBMS architecture, data models, schema types, and ER/EER design, including identification of entities and relationships in case-based scenarios.

Rubric Alignment Strategy

Criterion	Weight	How this submission targets Level 4 (Exemplary)
Concept Identification	30%	Explicitly identifies entities, subtypes, attributes, primary keys, constraints, weak entities, relationship types, DBA functions and relevant terminology.
Linkages	25%	Connects entities through named relationships, cardinalities and participation assumptions; shows inheritance and identifying relationships in diagrams.
Organization	20%	Each question follows the exact task order, uses structured tables and diagrams, and separates definitions, assumptions, analysis and final model.
Integration of Literature	15%	Includes a concise reference section and uses standard DBMS terminology consistent with established database design texts.
Clarity	10%	Uses consistent headings, tables, diagram captions, concise explanations, key legends and professional formatting.

Important modelling note

Where the assessment prompt does not explicitly provide an attribute, relationship or cardinality, the document labels it as a reasonable modelling assumption/proposed attribute rather than presenting it as a stated requirement. This keeps the answer complete while distinguishing prompt-derived facts from design choices.

Contents

1. Question 1 – Vehicle Management System
2. Question 2 – Bank Management System
3. Question 3 – E-Commerce Website
4. Question 4 – Hospital Management System – EER Diagram
5. Question 5 – Database Administrator (DBA)
6. Rubric Coverage Checklist
7. References

Question 1 – Vehicle Management System

The assessment describes Vehicle as the common supertype and Car and Truck as specialized subtypes. The specialization is explicitly stated to be disjoint and total.

1.1 Supertype and Subtypes

Element	Role	Explanation
Vehicle	Supertype	Stores information common to every vehicle.
Car	Subtype	Inherits all Vehicle attributes and adds car-specific attributes.
Truck	Subtype	Inherits all Vehicle attributes and adds truck-specific attributes.

1.2 Common Attributes of Vehicle

Attribute	Meaning	Key / Role
Vehicle_ID	Unique identifier for each vehicle	Primary Key
Registration_Number	Vehicle registration identifier	Candidate/unique attribute
Manufacturer	Vehicle manufacturer	Common attribute
Model	Vehicle model	Common attribute
Year	Manufacturing/model year	Common attribute
Purchase_Price	Purchase price of vehicle	Common attribute

1.3 Specialized Attributes

Subtype	Specialized attributes
Car	Number_of_Doors; Fuel_Type; Seating_Capacity
Truck	Load_Capacity; Truck_Type; Number_of_Axles

1.4 Primary Key and Inheritance

Vehicle_ID is the primary key of Vehicle. In an EER specialization, the subtype entity does not need a separate independent identifier for the inherited identity; Vehicle_ID is inherited and acts as the subtype primary key/foreign-key reference to Vehicle.

1.5 ISA Specialization and Constraints

- ISA means “is-a”: every Car is a Vehicle and every Truck is a Vehicle.
- Disjoint (d): a vehicle can belong to only one subtype—Car or Truck. It cannot simultaneously be both.
- Total (double line): every Vehicle must belong to one of the defined subtypes. No unclassified Vehicle is permitted under the stated requirements.

1.6 Complete EER Diagram

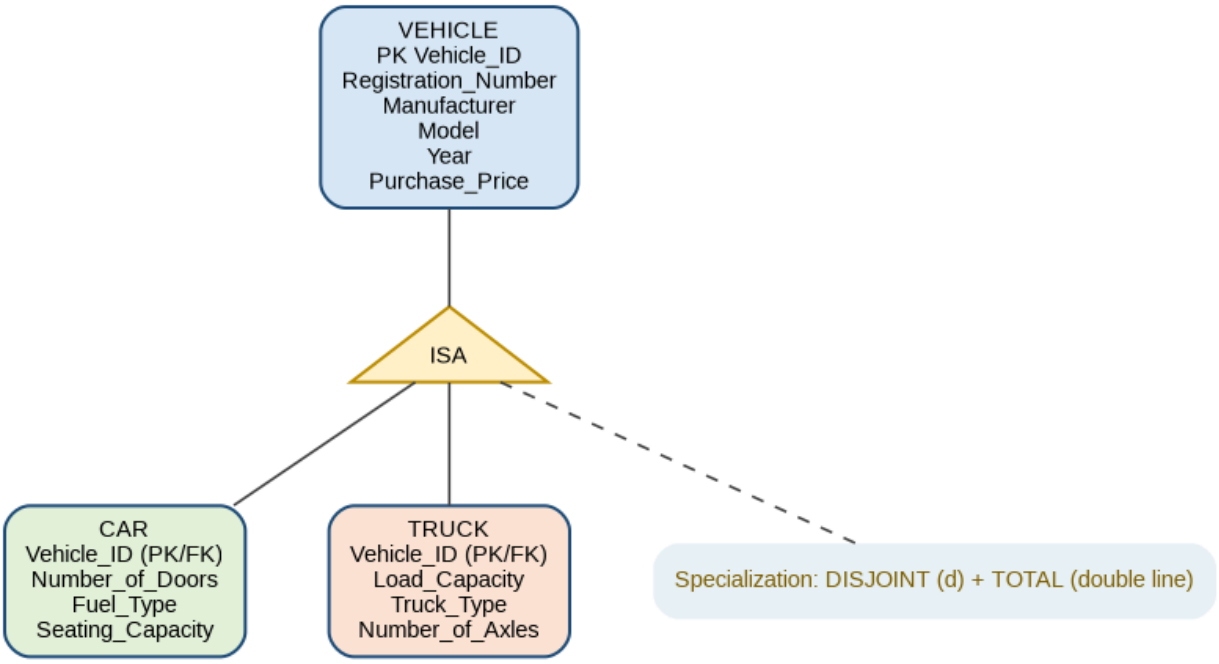


Figure 1. Vehicle supertype with Car and Truck subtypes; disjoint and total specialization.

1.7 Rubric-Focused Interpretation

Concept identification is complete because the supertype, subtypes, common attributes, specialized attributes, key and specialization constraints are explicitly stated. Linkages are represented through the ISA structure. The diagram integrates all requested elements in one model, satisfying the task's requirement for a complete EER representation. The assessment tasks explicitly require identification of the supertype/subtypes, common and specialized attributes, primary-key inheritance, ISA relationship, disjoint/overlapping choice, total/partial choice and a complete EER diagram.

Question 2 – Bank Management System

The prompt requires entities for customers, accounts, branches, employees, transactions, loans and loan payments, with attributes, keys, relationships, cardinalities and a complete ER diagram.

2.1 Major Entities and Attributes

Entity	Primary Key	Important Attributes
Customer	Customer_ID	Customer_ID
Account	Account_No	Account_Type, Balance, Opening_Date
Branch	Branch_ID	Branch_Name, Location, IFSC_Code
Employee	Employee_ID	Employee_Name, Designation, Salary
Transaction	Transaction_ID	Transaction_Date, Transaction_Type, Amount
Loan	Loan_ID	Loan_Type, Loan_Amount, Interest_Rate, Loan_Status
Loan_Payment	Payment_ID	Payment_Date, Payment_Amount

2.2 Relationships and Cardinalities

Relationship	Cardinality	Participation / Interpretation
Customer — holds — Account	1 : M	One customer may hold one or more accounts; each account belongs to a customer in the simplified model.
Account — maintained at — Branch	M : 1	Many accounts may be maintained by one branch; each account is maintained at a particular branch.
Branch — employs — Employee	1 : M	A branch employs several employees.
Customer — performs — Transaction	1 : M	A customer may perform multiple transactions.
Account — records — Transaction	1 : M	Proposed modelling relationship: transactions are recorded against an account.
Customer — applies for — Loan	1 : M	A customer may apply for multiple loans.
Loan — has — Loan_Payment	1 : M	A loan can have multiple payment records.

Modelling assumption for completeness

The prompt explicitly connects transactions to customers but does not state an Account–Transaction relationship. Because deposits and withdrawals normally affect an account, the diagram includes Account–Transaction as a proposed modelling relationship. If the faculty expects a strict prompt-only model, this relationship can be omitted without changing the explicitly required entities.

2.3 Entity Descriptions

- Customer: uniquely identifies the bank customer using Customer_ID.
- Account: stores account number, type, balance and opening date.
- Branch: represents the bank branch that maintains accounts and employs staff.
- Employee: records staff working at a branch.
- Transaction: captures deposit/withdrawal activity using a transaction identifier, date, type and amount.
- Loan: stores loan details and current loan status.
- Loan_Payment: records individual payments made against a loan.

2.4 Complete ER Diagram

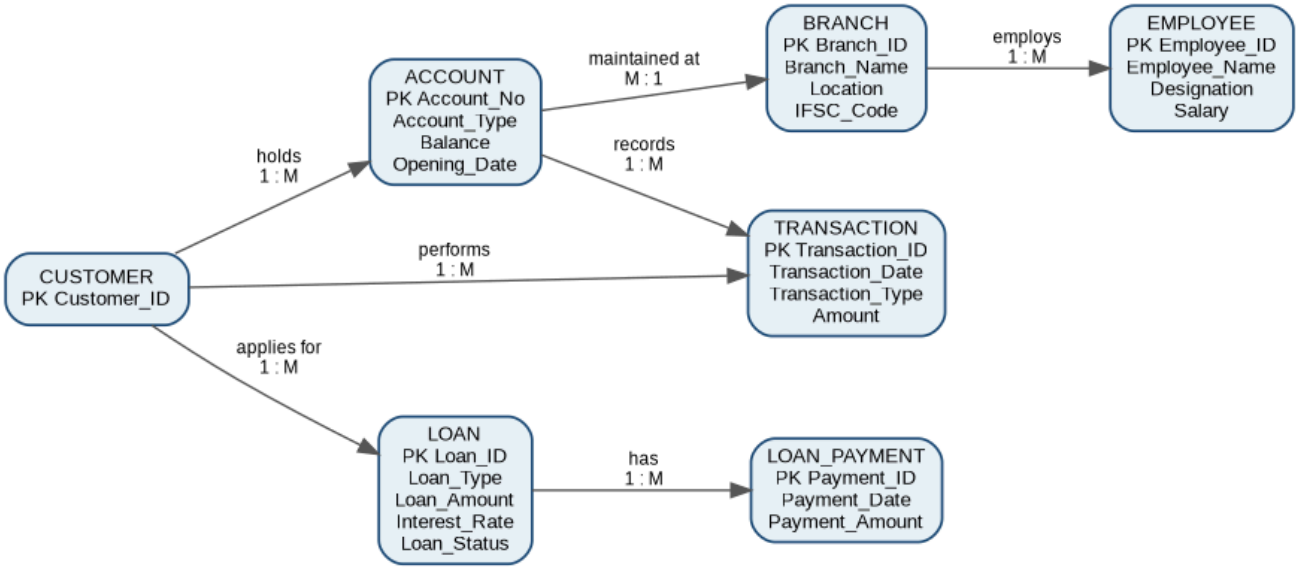


Figure 2. Bank Management System ER model with primary keys and relationship cardinalities.

2.5 Requirement Coverage

The assessment specifically asks for major entities, suitable attributes, primary keys, relationships, cardinalities and a complete ER diagram. All six task categories are addressed above, with the diagram consolidating the model.

Question 3 – E-Commerce Website

The assessment requires Customer, Product, Category, Order, Payment and Delivery; relationship attributes Quantity and Unit_Price on Order–Product; specialization/generalization for product categories; and cardinalities for the major relationships.

3.1 Major Entities and Primary Keys

Entity	Primary Key	Attributes
Customer	Customer_ID	Name, Email, Phone, Address
Product	Product_ID	Product_Name, Price, Stock_Quantity, Description
Category	Category_ID (proposed)	Category_Name (proposed)
Order	Order_ID	Order_Date, Order_Status, Total_Amount
Payment	Payment_ID	Payment_Date, Payment_Mode, Payment_Status
Delivery	Delivery_ID	Delivery_Date, Delivery_Status, Delivery_Address
Order_Item / associative entity	Order_ID + Product_ID	Quantity, Unit_Price

3.2 Order–Product Relationship

An order can contain multiple products, and a product can appear in multiple orders. Therefore, Order–Product is naturally modeled as an M:N relationship. The relationship attributes Quantity and Unit_Price are captured through the associative entity Order_Item, whose composite primary key is (Order_ID, Product_ID). This converts the M:N relationship into two 1:M relationships while preserving the required relationship attributes.

3.3 Product Specialization / Generalization

The prompt asks Product to be classified into suitable subtypes and to show inherited Product attributes. The following subtype design is proposed because the prompt names Electronics, Clothing and Grocery as example categories but does not supply subtype-specific attributes. Product is the supertype; Electronics, Clothing and Grocery are subtypes and inherit Product_ID, Product_Name, Price, Stock_Quantity and Description.

Subtype	Proposed specialized attributes	Inherited attributes
Electronics	Brand, Warranty_Period	All Product attributes
Clothing	Size, Material	All Product attributes
Grocery	Expiry_Date, Unit_Weight	All Product attributes

Specialization constraint recommendation

A disjoint specialization is the clearest default if each product is classified into exactly one of the named product categories. A total specialization is appropriate if the business rule requires every product to belong to one of the defined categories. Because the prompt asks the student to specify these constraints but does not explicitly state them, this is presented as a modelling recommendation rather than a quoted requirement.

3.4 Major Relationships and Cardinalities

Relationship	Cardinality	Reason
Customer — places — Order	1 : M	A customer can place multiple orders; each order is associated with the customer who placed it.

Relationship	Cardinality	Reason
Order — contains — Product	M : N	Each order can contain multiple products and products can occur in multiple orders; Quantity and Unit_Price belong to the associative relationship.
Order — has — Payment	1 : 1	Proposed default: one order has one payment record; this can be changed to 1:M if split/partial payments are permitted.
Order — has — Delivery	1 : 1	Proposed default: one order has one delivery record; business rules may permit multiple shipments.
Product — ISA — Electronics/Clothing/Grocery	1 : 1 subtype membership under disjoint total assumption	Subtypes inherit Product attributes.

3.5 Complete EER Diagram

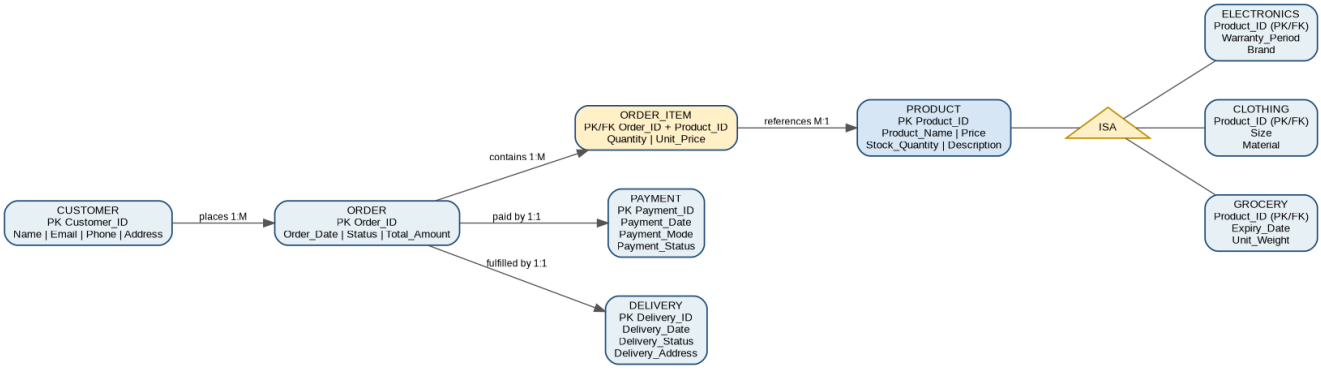


Figure 3. E-Commerce EER model showing Order_Item relationship attributes and Product specialization.

3.6 Requirement Coverage

The task list requires the six major entities, attributes and keys, Order–Product relationship attributes, product specialization, inherited attributes, specialization constraints, Customer–Order, Order–Payment and Order–Delivery relationships, cardinalities, and a complete EER diagram. These requirements are addressed in Sections 3.1–3.5.

Question 4 – Hospital Management System (EER Diagram)

The assessment requires Staff specialization into Doctor, Nurse and Administrative_Staff; inherited staff attributes; a weak Medical_Record entity dependent on Patient; an identifying relationship and partial key; and relationships among Doctor, Patient, Appointment, Department and Treatment with cardinalities and participation constraints.

4.1 Major Entities

Entity	Primary Key / Identification	Key Attributes
Staff	Staff_ID	Name, Address, Phone, Salary
Doctor	Inherited Staff_ID	Specialization, License_Number
Nurse	Inherited Staff_ID	Qualification, Shift
Administrative_Staff	Inherited Staff_ID	Proposed: Role, Office_Location
Patient	Patient_ID	Name, Date_of_Birth, Gender, Contact_Number
Medical_Record	Patient_ID + Record_Date (identifying key)	Record_Date, Diagnosis, Treatment_Details
Department	Department_ID (proposed)	Department_Name, Location
Appointment	Appointment_ID (proposed)	Appointment_Date, Appointment_Time, Status
Treatment	Treatment_ID (proposed)	Treatment_Date, Treatment_Details

4.2 Staff Specialization and Inheritance

- Staff is the supertype containing Staff_ID, Name, Address, Phone and Salary.
- Doctor inherits Staff attributes and adds Specialization and License_Number.
- Nurse inherits Staff attributes and adds Qualification and Shift.
- Administrative_Staff inherits Staff attributes. The prompt does not specify its unique attributes, so Role and Office_Location are proposed attributes for completeness.
- The prompt does not explicitly state whether Staff specialization is disjoint/overlapping or total/partial. A reasonable default for staff roles is disjoint if an individual has exactly one staff role; participation can be made total if every Staff record must be one of the listed subtypes. These are modelling assumptions, not stated facts.

4.3 Weak Entity: Medical_Record

Medical_Record is explicitly described as a weak entity because it is dependent on a particular Patient for identification. Its partial key can be Record_Date if one record date is unique within a patient. The full identifying key is therefore (Patient_ID, Record_Date). The Patient–Medical_Record relationship is an identifying relationship, normally represented with a double diamond in classic EER notation; the weak entity is represented with a double rectangle.

4.4 Hospital Relationships and Cardinalities

Relationship	Cardinality	Participation / Explanation
Patient — identifies/owns — Medical_Record	1 : M	A patient may have multiple records; every weak record depends on a patient for identification. Patient participation is total for the existence of a Medical_Record.
Patient — books — Appointment	1 : M	A patient may have multiple appointments.
Doctor — attends — Appointment	1 : M	A doctor may attend multiple appointments; each appointment is associated with a doctor in the simplified model.
Doctor — assigned to — Department	M : 1	Several doctors may be assigned to one department.

Relationship	Cardinality	Participation / Explanation
Doctor — provides — Treatment	1 : M	A doctor can provide multiple treatments.
Patient — receives — Treatment	1 : M	A patient can receive multiple treatments.

4.5 Participation Constraints

- Medical_Record has total/existence-dependent participation in the identifying Patient relationship because a weak record cannot exist without its owner Patient.
- Appointment participation is assumed mandatory for both a stored appointment and its referenced Patient/Doctor in the simplified model.
- Doctor–Department is modeled as each Doctor being assigned to one Department; if hospital policy permits visiting doctors or multiple departments, the relationship can be changed to M:N.
- Treatment links a Patient and Doctor; the prompt states that doctors provide treatments to patients, so both sides participate in treatment records.

4.6 Complete EER Diagram

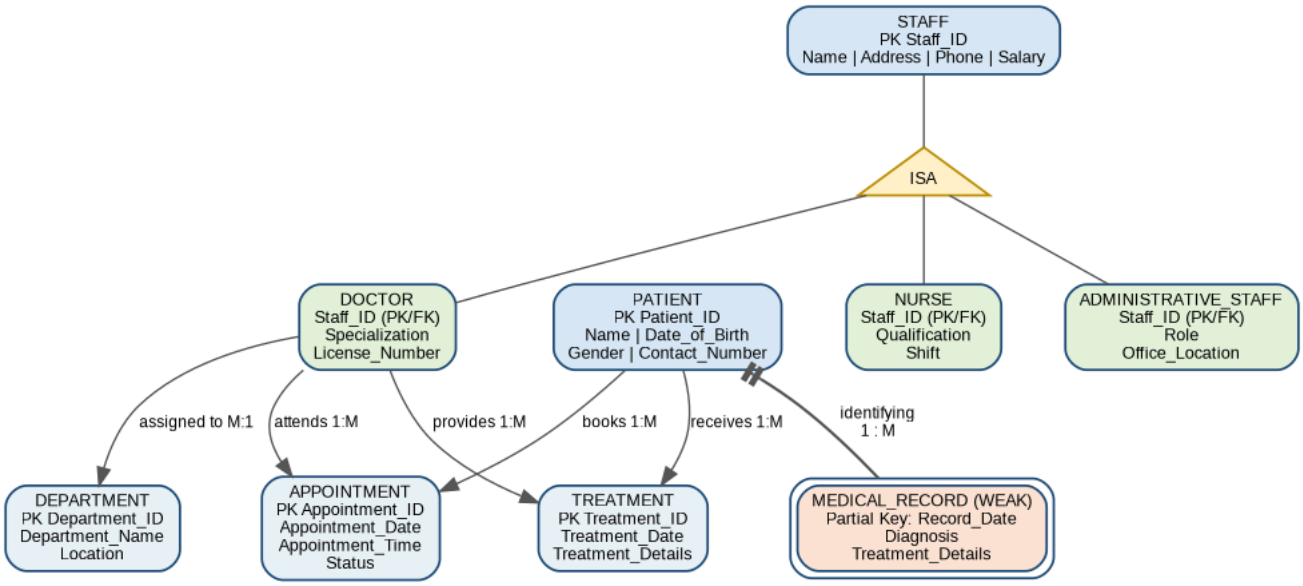


Figure 4. Hospital EER model showing Staff specialization and weak Medical_Record dependency.

4.7 Requirement Coverage

The task explicitly requires major entities, Staff specialization, inheritance, subtype attributes, weak-entity representation, partial key, identifying relationship, Doctor–Patient–Appointment–Department modelling, cardinalities and participation constraints. Sections 4.1–4.6 address each item in sequence.

Question 5 – Database Administrator (DBA)

The assessment describes the DBA as responsible for managing, maintaining, securing and ensuring reliable operation of organizational databases, including design, installation, access management, security, backup/recovery, performance monitoring, storage management and troubleshooting.

5.1 Definition and Core Role

A Database Administrator (DBA) is the professional responsible for the operational health, security, performance, availability and controlled use of an organization's database systems. The DBA ensures that organizational data remains accurate, consistent, protected and available to authorized users while supporting database design, maintenance, monitoring and recovery activities.

5.2 DBA Responsibilities – Detailed Answers to All Ten Tasks

Task	Answer
1. Role of a DBA	The DBA is the primary technical custodian of the database environment. The role combines administration, security, reliability, performance, maintenance and operational support.
2. Database design and implementation	The DBA participates in database architecture and implementation by helping define storage structures, database configuration, schemas, indexes, constraints and deployment standards. The DBA also installs and configures the DBMS and prepares environments for development, testing and production.
3. Database security and access control	The DBA protects data through authentication, authorization, roles, privileges, least-privilege access, auditing and security policies. Access is granted only to authorized users and applications, and sensitive operations can be monitored through audit trails.
4. Backup and recovery management	The DBA designs backup schedules and recovery procedures to protect against hardware failures, software errors, accidental deletion and operational disruptions. Recovery planning includes appropriate backup copies, retention, restoration testing and recovery procedures.
5. Performance monitoring and optimization	The DBA monitors slow queries, resource consumption, locking/contention, storage growth and other performance indicators. Optimization can include query/index analysis, configuration tuning, statistics maintenance and capacity planning.
6. Data integrity and consistency management	The DBA maintains reliable data by enforcing constraints, transaction controls, referential integrity, validation rules and appropriate concurrency/recovery mechanisms. The objective is to prevent invalid or contradictory database states.
7. Storage and database maintenance	The DBA manages storage allocation, database growth, indexes, tablespaces/files where applicable, routine maintenance, statistics and housekeeping activities. Maintenance reduces fragmentation and prevents storage-related outages.
8. Troubleshooting and disaster recovery	The DBA diagnoses database failures, connection problems, deadlocks, performance incidents, storage limits and corruption-related issues. Disaster recovery planning establishes procedures to restore services and data after major disruptions.
9. User and privilege management	The DBA creates or manages database accounts, roles and privileges, reviews access rights, removes unnecessary permissions and supports segregation of duties. Privileges should be aligned with job responsibilities.
10. Availability, reliability and security	The DBA combines monitoring, redundancy where applicable, backups, recovery procedures, security controls, maintenance and capacity planning to keep the database dependable and accessible to authorized users.

5.3 DBA Responsibility Matrix

Area	Primary Objective	Typical DBA Activities	Business Value
Design & Implementation	Build a sound database environment	Schema/configuration planning, installation, configuration	Stable foundation
Security	Protect information	Authentication, roles, privileges, auditing	Confidentiality and controlled access
Backup & Recovery	Prevent permanent data loss	Backups, restore tests, recovery procedures	Business continuity
Performance	Keep workloads efficient	Monitoring, query/index tuning, capacity planning	Fast and predictable systems

Area	Primary Objective	Typical DBA Activities	Business Value
Integrity	Keep data correct	Constraints, transactions, validation, consistency checks	Trustworthy information
Maintenance	Keep database healthy	Storage/index/statistics maintenance	Reduced operational risk
Troubleshooting	Resolve incidents	Root-cause analysis, diagnostics, corrective actions	Lower downtime
Availability	Keep services accessible	Monitoring, redundancy/recovery planning	Reliable operations

5.4 Integrated Explanation

A DBA's responsibilities are interconnected. Security controls who can use data; integrity controls whether data remains valid; performance management keeps access efficient; backup and recovery protect data when failures occur; storage and maintenance preserve operational health; and monitoring connects all of these activities by providing evidence of database condition. The assessment prompt emphasizes that the DBA must ensure data is stored efficiently, remains accurate and consistent, and is available to authorized users, while also addressing design, security, backup/recovery, performance, storage and troubleshooting.

5.5 Level-4 Takeaway

Exemplary-answer summary

A strong DBA answer should not treat these responsibilities as isolated definitions. It should show how design, security, integrity, performance, storage, backup/recovery, troubleshooting and access management collectively support availability, reliability and secure use of organizational data.

Rubric Coverage Checklist

Rubric area	Evidence in submission	Status
Concept Identification	All required entities, attributes, keys, subtypes, weak entity, partial key, ISA concepts, cardinalities and DBA responsibilities are explicitly identified.	✓
Linkages	Relationships are named and linked with cardinalities; inheritance is shown; weak-entity ownership and associative Order_Item modelling are explained.	✓
Organization	Five questions follow the assessment order, with numbered subsections, tables, diagrams and requirement-coverage notes.	✓
Integration of Literature	Reference section provides standard database-design texts suitable for supporting ER/EER and DBA terminology.	✓
Clarity	Consistent formatting, captions, key labels, modelling assumptions and concise explanatory paragraphs are used.	✓
Assessment task coverage	Every listed task under Questions 1–5 is directly addressed.	✓

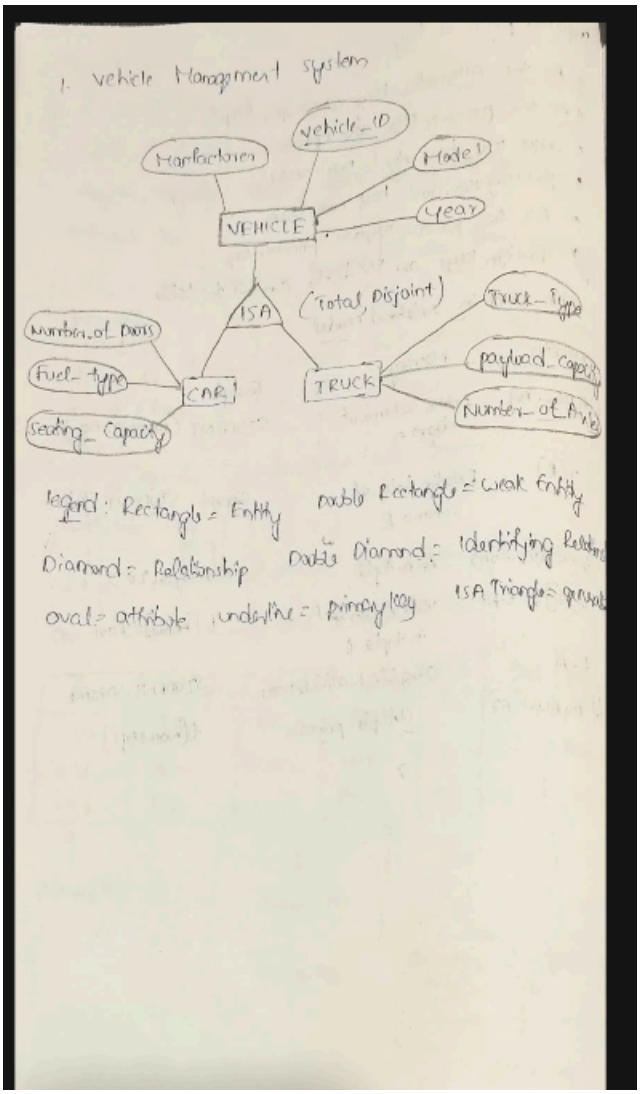
Submission Readiness Checklist

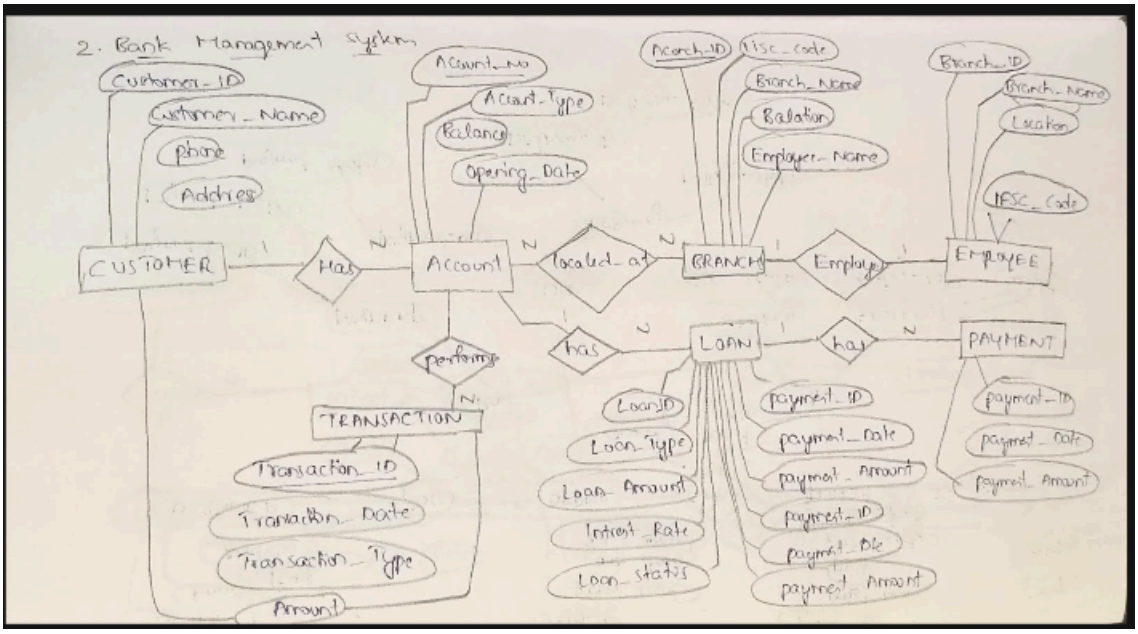
- Replace [ENTER STUDENT NAME] and [ENTER REGISTER NUMBER] on the cover and declaration.
- Add the student signature and submission date.
- Review proposed modelling assumptions with the course faculty if the instructor has a preferred convention for cardinalities or specialization constraints.
- Before submission, ensure the diagrams render clearly when printed or exported to PDF.
- If required by the institution, add the student's prescribed citation style or additional faculty-provided references.

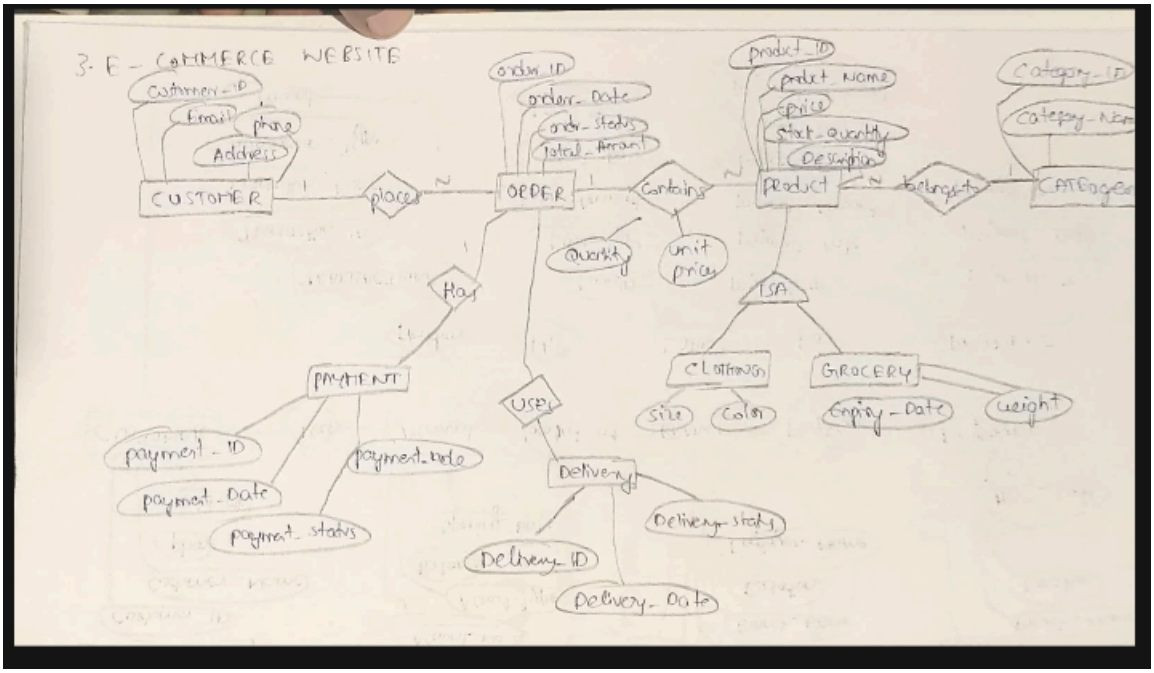
References / Suggested Literature

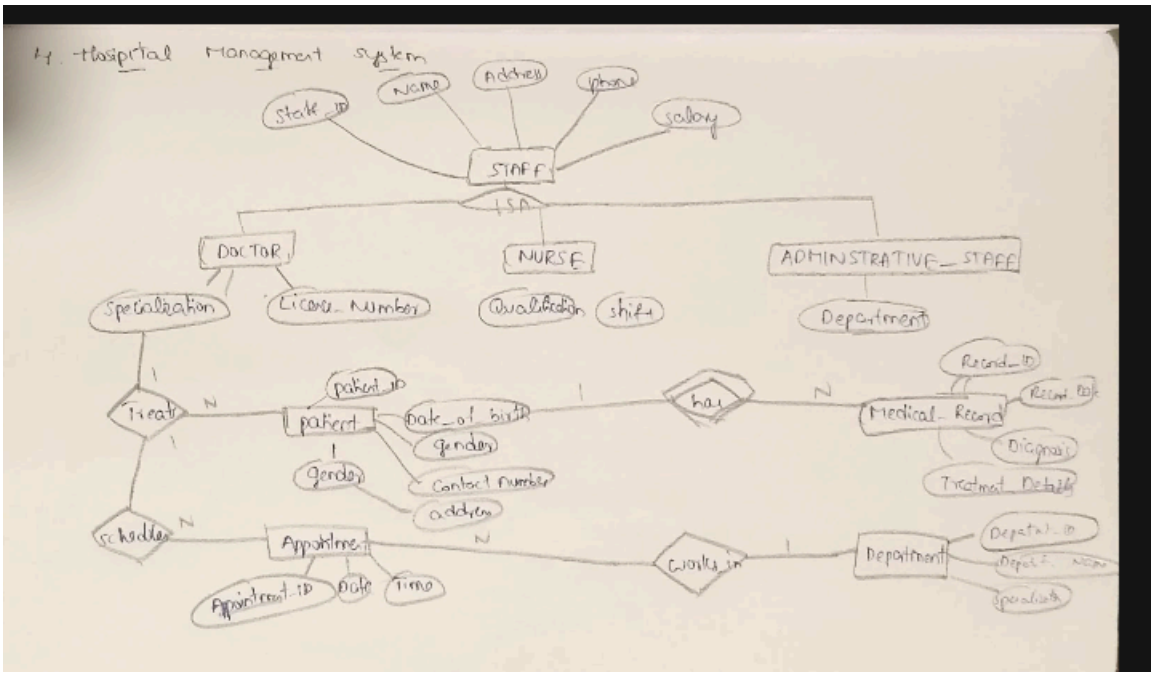
Source	Reference	Use in this submission
Elmasri, R., & Navathe, S. B.	Fundamentals of Database Systems. Pearson.	Useful for ER/EER modelling, specialization/generalization, weak entities, keys and relational database concepts.
Silberschatz, A., Korth, H. F., & Sudarshan, S.	Database System Concepts. McGraw-Hill.	Useful for database architecture, relational concepts, transactions, recovery, security and administration.
Assessment Brief – SIMATS Engineering	Database Management Systems, CSA05, Continuous Internal Assessment – Short Answer Text.	Primary task source for the five questions, CO1 and the grading rubric.

Note: The first two references are suggested standard literature for strengthening the submission's academic framing; the assessment brief remains the controlling source for the exact question requirements and rubric.


1. Vehicle Management system
Vehicle_ID
Manfacturer
Model
Year
VEHICLE
ISA
(total, Disjoint)
Truck_Type
payload_capacity
Number_of_Axle
TRUCK
Number_of_Doors
Fuel_type
CAR
Seating_Capacity
legend: Rectangle = Entity
Double Rectangle = weak Entity
Diamond = Relationship
Double Diamond = Identifying Relationship
oval = attribute
underline = primary key
ISA Triangle = generalization


2. Bank Management system
Customer_ID
Customer_Name
Phone
Address
Account_No
Account_Type
Balance
Opening_Date
Branch_ID
Bisc_code
Branch_Name
Balance
Employer_Name
Branch_ID
Branch_Name
Location
IFSC_Code
CUSTOMER
Has
Account
located_at
BRANCH
Employs
EMPLOYEE
performs
has
LOAN
pay
PAYMENT
TRANSACTION
Transaction_ID
Transaction_Date
Transaction_Type
Amount
LoanID
Loan_Type
Loan_Amount
Intrest_Rate
Loan_status
payment_ID
payment_Date
payment_Amount
payment_ID
payment_Date
payment_Amount


3. E - COMMERCE WEBSITE
Customer_ID
Email
phone
Address
CUSTOMER
places
ORDER
order_ID
order_Date
order_status
total_Amount
Contains
Quantity
unit price
product_ID
product_Name
price
stock_quantity
Description
Product
belongs to
Category_ID
Category_Name
CATEGORIES
Has
PAYMENT
payment_ID
payment_Date
payment_status
payment_mode
USES
Delivery
Delivery_ID
Delivery_Date
Delivery_status
ISA
CLOTHINGS
Size
Color
GROCERY
Expiry_Date
weight


4. Hospital Management System
Staff_ID
Name
Address
phone
salary
STAFF
ISA
DOCTOR
NURSE
ADMINSTRATIVE_STAFF
Specialization
License_Number
Qualification
shift
Department
Treat
patient_ID
patient
Date_of_birth
Gender
Contact number
address
gender
has
Record_ID
Record_Date
Medical_Record
Diagnosis
Treatment_Details
schedules
Appointment
Appointment_ID
Date
Time
works_in
Department
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Department_Name
Specialization